

System for High Integrity Elimination of Low-Altitude Drones (SHIELD)

Milestone 1 Progress Evaluation

1. Project Title, Team Members

System for High Integrity Elimination of Low-Altitude Drones (SHIELD)

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2. Faculty Advisor

Madhur Tiwari, PhD - mtiwari@fit.edu

3. Client

Firat Irmak, PhD - Aerospace Engineering Department, Florida Institute of Technology

4. Progress of current Milestone

Task	Completion %	Bryson Noble	TODO
Compare and select technical tools	100%	100%	none
"Hello world" demos	100%	100%	none
Resolve technical challenges	90%	90%	Get access to AI.Panther to train a better model.
Compare and select collaboration tools	100%	100%	none
Requirement Document	100%	100%	none
Design Document	100%	100%	none
Test Plan	100%	100%	none

5. Discussion of each accomplished task (and obstacles) for the current Milestone:

- **Compare and select technical tools:** selected OpenCV for camera capture and preprocessing, YOLO26s for object detection, ByteTrack for multi-object tracking and ID assignment, a Kalman filter for state estimation and trajectory prediction, Unity to create a virtual camera to run tests without a physical camera, STE (SHIELD Test

Environment), an in-house C#/VB.NET to run automated tests and verify requirements. We originally planned to use a Raspberry Pi 5 with a Hailo-9 AI HAT+, but after doing further research, we switched to the Jetson Orin Nano. The Orin gives us sufficient computing power while remaining within our allocated budget. To compare detection models, I built a weighted decision matrix comparing YOLOv8, v11, and v26 in nano/small/medium sizes on accuracy and inference speed. I selected YOLO26s as the best balance of speed and accuracy while maintaining a framerate of above 30 fps. A few obstacles we are yet to face are potential thermal throttling and additional latency that could affect our frame rate.

- **"Hello world" demos:** I built the CV/detection pipeline and the GCS executable for ground control. The demo uses our YOLO model, ByteTrack tracking, and Kalman smoothing using the Unity virtual camera as input. This demo showed that all of our software works together smoothly before we begin testing and implementing on hardware.
- **Resolve technical challenges:** The main remaining challenge is training compute. My GTX 1660 is too slow and lacks sufficient VRAM to train a YOLO26s model with a batch size of 64. I need to submit a proposal to Dr. Irmak, explaining why I need access to AI.Panther so that I can train our final model.
- **Compare and select collaboration tools:** I use Visual Studio for development, Teams for communication and document sharing with other team members, and GitHub as my VCS.
- **Requirement Document:** The requirement document includes only requirements from my subsystem (Avionics & Software) and not the entire SHIELD project, as that is outside my scope as the individual submitter for this SWE senior design class. I authored the AVS requirements with help from Tharushi Pelendagama. Each requirement traces to a parent system requirement and includes a cited rationale and planned verification method.
- **Design Document:** I have defined the software and hardware architecture for the detection/tracking portion of the project. The camera and edge computer have been researched and selected. The software pipeline is also defined: camera → OpenCV capture/preprocessing → YOLO26s detection → ByteTrack tracking → Kalman filter state estimation → publish to GNC at 30 Hz or faster. There will also be a heartbeat system implemented between the AVS and the GNC HW/SW, as well as between the GNC HW/SW and the Ground Control System. The color scheme for the GCS is also based on research around chromatic vibration; I seek to make the GCS as easy to see and easy on the eyes as possible.
- **Test Plan:** SW requirements will be verified by the STE test harness and test scripts. I will be tracing as many requirements as possible to these test scripts to provide additional verification methods and ease the physical testing/analysis load. At the very least, every AVS requirement shall be verifiable through automated testing on both software and

hardware. Other requirements list flight testing and analysis as verification methods, but most of that work is outside my scope in this project.

6. Discussion of contribution of each team member to the current Milestone:

- Bryson Noble: As the Avionics and Software lead, I am responsible for all targeting and detection software on SHIELD. I selected the Jetson Orin Nano as our edge computer and YOLO26s as our object detection model. I am working on the test harness and detection/tracking pipeline, the AVS subsystem requirements, the software architecture and GCS GUI design, and the software test plan. On the team's recently-submitted updated SRR, I contributed to the software deliverables and AVS requirements sections.

7. Plan for the next Milestone

Task	Bryson Noble
Test the pipeline and model on Orin Nano and make necessary changes for implementation	Implement, run, collect data
Refine the model to improve accuracy, and begin training the final model to be used on Orin Nano	Collect data, train, evaluate
Add moving targets to the STE test library to allow for more robust automated testing	Implement
Continue development on GCS, so it is ready to control drone hardware once prototyping begins	Design, implement

8. Discussion of each planned task for the next Milestone:

- **Test pipeline....:** Test everything I've done and have gotten to work on emulation on the actual hardware, now that we have the Orin Nano checked out in the HSDC. This will allow me to get ahead of problems and resolve any implementation issues before it is time to mount it on the drone, after which it will be more difficult to make changes.
- **Refine model....:** Find the best combination of datasets that works for our detection needs, so I want to use different combinations of data to create the best dataset for our project using small batch sizes to train and benchmark each model, then once I have the best-performing model, I will train YOLO26s at batch size 64 using AI.Panther.
- **Add moving....:** Add moving targets to the STE test library so that test scripts can spawn them, allowing me to test our detection and tracking requirements more thoroughly.
- **Continue development....:** Create GCS GUI and ensure that engage/abort commands work. Implement all items listed in the risk management table. Verify wireless communication with the hardware if implemented during this milestone.

9. Date(s) of meeting(s) with Client during the current milestone:

- 08/31/2026
- 09/14/2026
- 09/28/2026

10. Client feedback on the current milestone

- 08/31: Test the drone against a balloon at first. Progress from stationary to vertical, then lateral motion. Keep trials as consistent as possible.
- 09/14: Do not spend ~\$1300 on the VOXL 2 Mini. Just use Orin Nano and make it fit in the drone. The group cannot replace VOXL if it breaks.
- 09/14: Camera must run at 30 fps or faster.
- 09/14: Justify the 30-frame lost-target requirement.
- 09/14: Autonomous chasing may be difficult; it depends on the onboard computer's capabilities.
- 09/28: Operator needs a button to shut down the drone or land immediately. Implement in GCS.

11. Date(s) of meeting(s) with Faculty Advisor during the current milestone:

- 09/16/2026

12. Faculty Advisor feedback on the current milestone

- Develop GNC and detection separately with a plan to integrate at the end of the project.
- Do more research on classical computer vision for object detection, such as template matching.

Signature: _____ Date: _____

13. Evaluation by Faculty Advisor

- Faculty Advisor: detach and return this page to Dr. Chan (HC 209) or email the scores to pkc@cs.fit.edu
- Score (0-10) for each member: circle a score (or circle two adjacent scores for .25 or write down a real number between 0 and 10)

Bryson	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Noble																

Signature: _____ Date: _____